WEEK-VIII

TYPES OF MODELING

$\{(x, y) \mid x \in X \;\&\; y \in Y\}$

feature space

label space

DISTRIBUTION D OVER $X \times Y$

cross

FEATURE SPACE $\in \mathbb{R}^d$

$\{+1, -1\}$

$\{x_1, \ldots, x_n\}$
$\{y_1, \ldots, y_n\}$

(I) UNKNOWN BUT FIXED

$(x_i, y_i) \sim D$

Data-generating probabilistic mechanism

LINK BETWEEN TRAIN and TEST.

Both the training and the test dataset will come from the same underlying distribution **D**

CLASSIFICATION (Supervised Learning)

(II) structural assumptions

- GENERATIVE MODEL
- DISCRIMINATIVE MODEL

modelling

GENERATIVE MODEL

modelling feature generation also

MODEL

- $P(x, y)$

joint

$P(x) \cdot P(y/x)$

↳ NEXT

DISCRIMINATIVE MODEL

Deterministic mapping

MODEL

- $P(y/x)$

conditional

Eg: K-NN
DECISION-TREES

↳ $P(y=1/x) = 1$ if decision tree for x says 1.

$P(y=1/x) = 1$ if majority of neighbours say 1
$= 0$ otherwise

trivial Probability

Q. Which one is better?

* Knowledge of how the data is generated helps us discriminate between the classes.

DATA = $\{ (x_1, y_1) \ldots (x_n, y_n) \}$

$x_i \in \{0,1\}^d \longrightarrow$ Binary features

$y_i \in \{0,1\}$

EXAMPLE : SPAM CLASSIFICATION

$y = \{0, 1\}$ (0 → Non-spam, 1 → Spam)

$x_i \in \{0,1\}^d$ ← # words in dictionary

Eg: "Hello, how are you?"

→ $\begin{array}{ccccccccccc} \text{About} & & & & \text{ARE} & \text{HELLO} & & \text{HOW} & & \text{YOU} & \text{ZEBRA} \\ 0 & 0 & 0 & \cdots & 1 & 1 & \cdots & 1 & \cdots & 1 & 0 \end{array} \in \mathbb{R}^d$

$P(x, y)$ ("Hello, how are you?", spam) = 0.01

$$P(x,y) = \underbrace{P(x)}_{P(\text{email})} \cdot \underbrace{P(y/x)}_{P(\text{spam}/\text{email})} = \underbrace{P(y)}_{P(\text{spam})} \cdot \underbrace{P(x/y)}_{P(\text{email}/\text{spam})}$$

modelling the structure of an email is not necessary; one can choose a Discriminative model to model p(spam/email)

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we care about p(spam/email) only to make prediction.

likelihood of seeing a spam or a non-spam

structure of a particular email given spam.

GENERATIVE STORY

STEP-1 : DECIDE the Label by tossing a coin

$P(y_i = 1) = p$

$P(y_i = 0) = (1 - p)$

STEP 2 : DECIDE Features using the Label in step 1 by $P(x_i / y_i)$

d=3

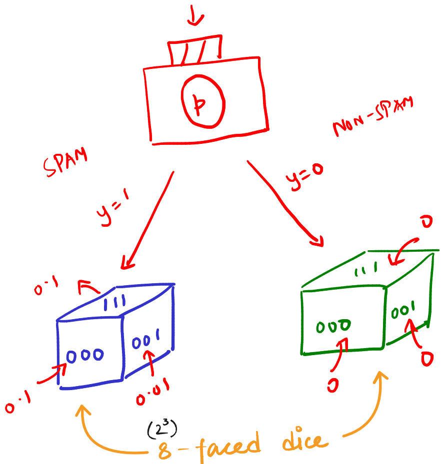
	W_1	W_2	W_3	
mutually exclusive events	0	0	0	0.1
	0	0	1	0.01
	0	1	0	0.02
	0	1	1	0.03
	1	0	0	0.04
	1	0	1	0.5
	1	1	0	0.2
	1	1	1	0.1

$\Sigma p = 1$

$2^3 = 8$ different feature vectors as d = 3

* features are not independant of each other.

don't need this element's probability as the total probability is 1.



faces are not equally likely

	W_1	W_2	W_3	
mutually exclusive events	0	0	0	0
	0	0	1	0
	0	1	0	0
	0	1	1	0.5
	1	0	0	0.5
	1	0	1	0
	1	1	0	0
	1	1	1	0

$\Sigma p = 1$

$2^3 = 8$ different feature vectors as d = 3

free parameters

\# parameters in the model $= \underbrace{1}_{\text{to decide label}} + \underbrace{(2^d - 1)}_{P(x/y=1)} + \underbrace{(2^d - 1)}_{P(x/y=0)}$

$= 1 + 2(2^d - 1)$

$= 2^{d+1} - 1$

- **Multiclass Classification Problem:**

K- classes - $y_i \in \{1, 2, 3, \dots, K\}$

d - binary features - $x_i \in \{0, 1\}^d$

\# parameters to be estimated = parameters in the model

$= (K-1) + K(2^d - 1)$

$= \boxed{K(2^d) - 1}$

ISSUE

- Too many parameters!
- Not a reasonable story.

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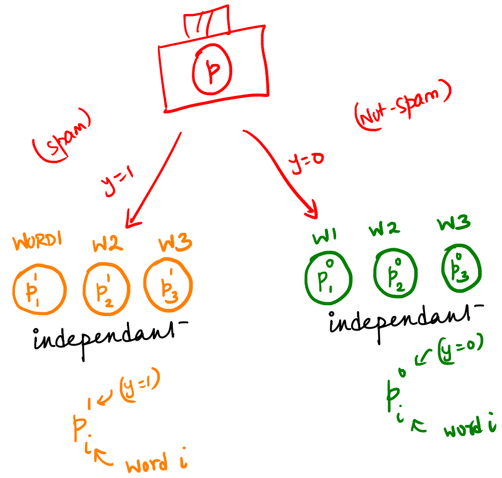
d = 3

SPAM

0 0 0
0 0 1
⋮
0 1 1
⋮
1 1 1

0 1 1
TAIL HEAD HEAD

d - coins:



NON-SPAM

0 0 0
0 0 1
⋮
1 1 1

2^d

: d - coins

CLASS CONDITIONAL INDEPENDENCE ASSUMPTION!

\# parameters $= 1 + d + d = \underbrace{2d + 1}_{\text{MANAGEBLE}}$

\# parameters to be estimated = parameters in the model

$= \boxed{(K-1) + Kd}$

Example

$y=1$

$p_1^1 \quad p_2^1 \quad p_3^1$

$$p\left(x = [0 \;\; 1 \;\; 0] / y=1\right)$$
$$= (1-p_1^1)\cdot(p_2^1)(1-p_3^1)$$

STEP 1 : $P(y=1) = p$

$P(y=0) = 1-p$

STEP 2 : $p\left(x = [f_1 \;\; f_2 \; \cdots \; f_d] / y\right)$; $f_i \in \{0,1\}$, $y_i \in \{0,1\}$

$$= \underbrace{\prod_{i=1}^{d} \left(p_i^y\right)^{f_i} \left(1 - p_i^y\right)^{(1-f_i)}}$$

iid

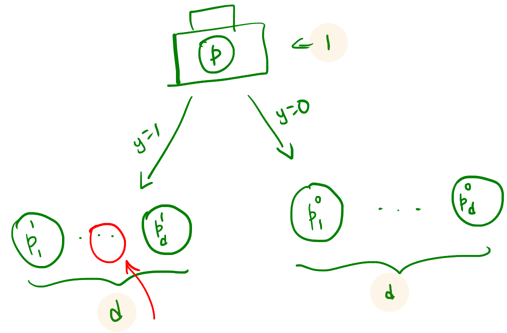
FEATURES ARE CONDITIONALLY INDEPENDENT GIVEN LABEL

- How to estimate the parameters
- SOLUTION : MAXIMUM LIKELIHOOD !

⭐ The primary goal of a generative model is to find the parameters that maximize the probability (likelihood) of the training data occurring.

Calculating joint probabilities requires multiplying many tiny fractions, which causes computational underflow in computers. Taking the logarithm solves this by converting multiplication into addition, keeping the math numerically stable without changing the final optimal result.

Standard machine learning algorithms are built to minimise an Objective (cost/loss) function, not maximise it. By taking the negative of the log-likelihood, we perfectly flip our maximisation goal into a standard minimisation problem.



CLASS CONDITIONAL INDEPENDENCE

$2d+1$ parameters in the model

PARAMETER ESTIMATION

$p, \{ p_1^1, \ldots, p_d^1 \}, \{ p_1^0, \ldots, p_d^0 \}$

label (superscript), feature (subscript), d

MAX. LIKELIHOOD ESTIMATES

$\hat{p}$ is trying to estimate the probability wheather an email is spam or not.

1 parameter $\rightarrow$ $\hat{p} = \frac{1}{n}\sum_{i=1}^{n} y_i$ $\rightarrow$ {Fraction of spam emails in the dataset}

sample mean of the labels

(y = 1)-labelled datapoints

2d parameters $\rightarrow$ $\forall j \in \{1, \ldots, d\}$, $\forall y \in \{0,1\}$, $\forall i \in \{1, \ldots, n\}$

$$\hat{p}_j^{\,y} = \frac{\sum_{i=1}^{n} \mathbb{1}\left(f_j^i = 1, y_i = y\right)}{\sum_{i=1}^{n} \mathbb{1}\left(y_i = y\right)}$$

presence of j^{th} word in the email

j^{th} feature

i^{th} datapoint

Number of emails with label y.

datapoints

$\rightarrow$ Fraction of y-labelled emails that contain the j^{th} word.

datapoints

j^{th} feature

PREDICTION

Given $x^{test} \in \{0,1\}^d$, What is $\hat{y}^{test}$?

$$P\left(y^{test} = 1 \,/\, x^{test}\right) > P\left(y^{test} = 0 \,/\, x^{test}\right)$$

$$\Rightarrow \hat{y}^{test} = 1$$

$= 0$ otherwise.

How to obtain $P(y/x)$ from $P(y)$ and $P(x/y)$?

BAYES' RULE ! $\rightarrow P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$

$$P(y^{test}=1 \mid x^{test}) = \frac{P(x^{test}/y^{test}=1) \cdot P(y^{test}=1)}{P(x^{test})}$$

$P(x^{test}) \rightarrow$ +ve denominator does not affect comparison.

$$P(y^{test}=0 \mid x^{test}) = \frac{P(x^{test}/y^{test}=0) \cdot P(y^{test}=0)}{P(x^{test})}$$

$P(x^{test}) \rightarrow$ +ve denominator does not affect comparison

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$$\underbrace{P(x^{test}/y^{test}=1)} \cdot P(y^{test}=1)$$

• features are independant of each other

$$= P(x^{test}=[f_1\ f_2 \dots f_d]/y^{test}=1) \cdot P(y^{test}=1)$$

using MLE here $\rightarrow$

$$= \left(\prod_{j=1}^{d} (\hat{p}_j^{1})^{f_j} (1-\hat{p}_j^{1})^{(1-f_j)}\right) \cdot \hat{p}$$

d - parameters + 1 - parameter

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if

$$X_{test} = \begin{array}{cccc} f_1 & f_2 & f_3 & f_4 \\ \left[\begin{matrix} 0 & 0 & 1 & 0 \end{matrix}\right. & & & \left.\right] \end{array}$$

$$\left(\prod_{j=1}^{d} (\hat{p}_j^{1})^{f_j} (1-\hat{p}_j^{1})^{(1-f_j)}\right) \cdot \hat{p} > \left(\prod_{j=1}^{d} (\hat{p}_j^{0})^{f_j} (1-\hat{p}_j^{0})^{(1-f_j)}\right) \cdot (1-\hat{p})$$

$\Rightarrow$ PREDICT $\hat{y}^{test} = 1$

else $\hat{y}^{test} = 0$.

NAIVE BAYES ALGORITHM.

MODEL uses 2 main Things (assumptions)

• CLASS CONDITIONAL INDEPENDENCE (iid) $\rightarrow$ features are independant given the labels

$2^d \rightarrow 2d$

• $\boxed{\text{BAYES'}}$ THEOREM

• may not hold in practice

• $\boxed{\text{NAIVE}}$ Assumption

• Still works well in practice, especially when d is large.

PITFALLS IN NAIVE BAYES TO WATCH OUT FOR.

- If a word does not appear in the train set but appears in a test datapoint,

$\hat{p}^1_j = 0 \quad \hat{p}^0_j = 0$

$$P\left(y^{test}=1 \,\middle|\, \underset{\substack{\text{zebra}\\=1}}{x^{test}=[f_1, f_2 \ldots f_d]}\right) \propto \underbrace{\left(\prod_{i=1}^{d} \underbrace{(\hat{p}^1_i)}_{0}^{\,f_i \to 1} \underbrace{(1-\hat{p}^1_i)}_{1}^{(1-f_i)\to 0}\right)\hat{p}}_{0}$$

$$P\left(y^{test}=0 \,\middle|\, \underset{\substack{\text{zebra}\\=1}}{x^{test}=[f_1 \ldots f_d]}\right) \propto \underbrace{\left(\prod_{i=1}^{d} \underbrace{(\hat{p}^0_i)}_{0}^{\,f_i \to 1} \underbrace{(1-\hat{p}^0_i)}_{1}^{(1-f_i)}\right)(1-\hat{p})}_{0}$$

$P(y^{test}=0 / x^{test}) = P(y^{test}=1 / x^{test}) = 0$

Possible Fix

- Can add two "pseudo" datapoints / emails with all words present - one datapoint / email has label 0 and another has label 1

LAPLACE SMOOTHING

$$\begin{array}{cc} & y^{pseudo}_1 \\ x^{pseudo}_1 \; [1 \quad 1 \quad 1 \quad 1 \quad 1 \quad \cdots \quad 1] & 1 \\ x^{pseudo}_0 \; [1 \quad 1 \quad 1 \quad \cdots \quad 1] & 0 \end{array}$$

Every feature appears at least once in both the classes → Probability estimates from maximum likelihood cannot be zero.

DECISION FUNCTION OF NAIVE BAYES.

Given x_{test}; $y_{test}=1$ if $\dfrac{P(y_{test}=1 / x_{test})}{P(y_{test}=0 / x_{test})} \geq 1$

$$\log\left(\frac{P(y_{test}=1 \mid x_{test})}{P(y_{test}=0 \mid x_{test})}\right) \geq 0 \qquad (\log(1)=0)$$

$$\log\left(\frac{\underbrace{P(x_{test} \mid y_{test}=1)\cdot P(y_{test}=1)}\,/\,P(\cancel{x_{test}})}{\underbrace{P(x_{test} \mid y_{test}=0)\cdot P(y_{test}=0)}\,/\,P(\cancel{x_{test}})}\right) \geq 0$$

$x_{test} = [f_1\ f_2 \dots f_d]$

$$\log\left(\prod_{i=1}^{d} \frac{(\hat{p}_i^1)^{f_i}\,(1-\hat{p}_i^1)^{(1-f_i)} \cdot \hat{p}}{(\hat{p}_i^0)^{f_i}\,(1-\hat{p}_i^0)^{(1-f_i)}\,(1-\hat{p})}\right) \geq 0$$

$$\equiv \log\left(\prod_{i=1}^{d}\left(\frac{\hat{p}_i^1}{\hat{p}_i^0}\right)^{f_i}\left(\frac{1-\hat{p}_i^1}{1-\hat{p}_i^0}\right)^{(1-f_i)} \cdot \frac{\hat{p}}{1-\hat{p}}\right) \geq 0$$

$$\equiv \sum_{i=1}^{d}\left(f_i \underbrace{\log\left(\frac{\hat{p}_i^1}{\hat{p}_i^0}\right)} + (1-f_i)\underbrace{\log\left(\frac{1-\hat{p}_i^1}{1-\hat{p}_i^0}\right)}\right) + \log\left(\frac{\hat{p}}{1-\hat{p}}\right) \geq 0$$

$$\boxed{\log\frac{m}{n} = \log m - \log n}$$

$x_{test} = [f_1 \dots f_d]$

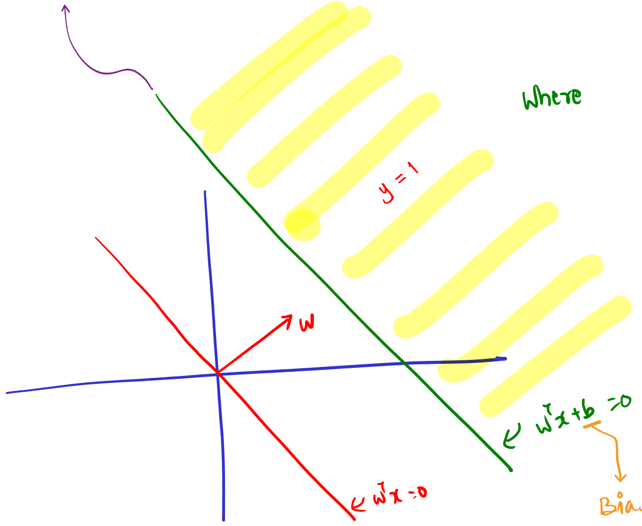
$$\equiv \underbrace{\sum_{i=1}^{d} f_i\left(\log\left(\frac{\hat{p}_i^1(1-\hat{p}_i^0)}{\hat{p}_i^0(1-\hat{p}_i^1)}\right)\right)} + \underbrace{\log\left(\frac{(1-\hat{p}_i^1)}{(1-\hat{p}_i^0)}\right) + \log\left(\frac{\hat{p}}{1-\hat{p}}\right)} \geq 0$$

DECISION FUNCTION is of the form

Predict $y_{test} = 1$ if $\underset{\in \mathbb{R}^d}{w^T} x_{test} + b \geq 0$

where $w_i = \underbrace{\log\left(\frac{\hat{p}_i^1(1-\hat{p}_i^0)}{\hat{p}_i^0(1-\hat{p}_i^1)}\right)}$, $b =$

Linear Decision Boundary



$\{x : P(y=0 \mid x) = P(y=1 \mid x)\}$

Bias: how much away from the origin

CONCLUSION:

→ DECISION FUNCTION OF NAIVE BAYES IS LINEAR!

DATA : $\{ (x_1, y_1), \ldots \quad (x_n, y_n) \}$

$x_i \in \mathbb{R}^d \qquad y_i \in \{0, 1\}$

A GENERATIVE STORY

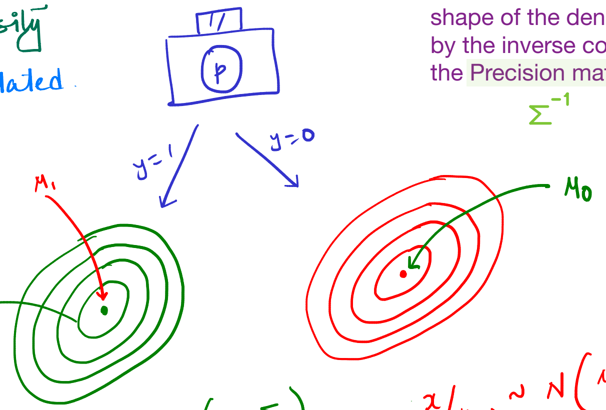
Σ → tells us how the density contours look like.

density ↘ regions w/ same density

Σ → tells us how features are co-related.

In a multivariate Gaussian distribution, the shape of the density contours is controlled by the inverse covariance matrix, also called the Precision matrix.

Σ^{-1}



PARAMETERS

- p

ellipses around the mean and the direction of these ellipses is determined by Σ.

- μ_0, μ_1

- Σ : co-variance matrix, common for both classes

$x/_{y=1} \sim N(\mu_1, \Sigma)$ $\qquad$ $x/_{y=0} \sim N(\mu_0, \Sigma)$

NOTE : In this model, covariances are assumed to be same

MAXIMUM LIKELIHOOD ESTIMATES

$$\hat{p} = \frac{\sum_{i=1}^{n} y_i}{n}$$

← FRACTION OF points labelled 1.

$$\hat{\mu}_1 = \frac{\sum_{i=1}^{n} \mathbb{1}(y_i = 1) \cdot x_i}{\sum_{i=1}^{n} \mathbb{1}(y_i = 1)}$$

← Sample mean of data points labelled 1.

$$\hat{\mu}_0 = \frac{\sum_{i=1}^{n} \mathbb{1}(y_i = 0) \cdot x_i}{\sum_{i=1}^{n} \mathbb{1}(y_i = 0)}$$

← Sample mean of data points labelled 0.

$$\hat{\Sigma} = \frac{1}{n} \sum_{i=1}^{n} (x_i - \hat{\mu}_{y_i})(x_i - \hat{\mu}_{y_i})^T$$

y_i is the label associated with x_i

PREDICTION ? Bayes rule.

$$P(y_{test} \mid x_{test}) \propto \underbrace{P(x_{test} \mid y_{test})}_{f(x_{test};\, \hat{\mu}_{y_{test}}, \hat{\Sigma})} \cdot \underbrace{P(y_{test})}_{\hat{p}}$$

Predict $y_{test} = 1$ if

$$f(x_{test};\, \hat{\mu}_1, \hat{\Sigma}) \cdot \hat{p} \geq f(x_{test};\, \hat{\mu}_0, \hat{\Sigma}) \cdot (1-\hat{p})$$

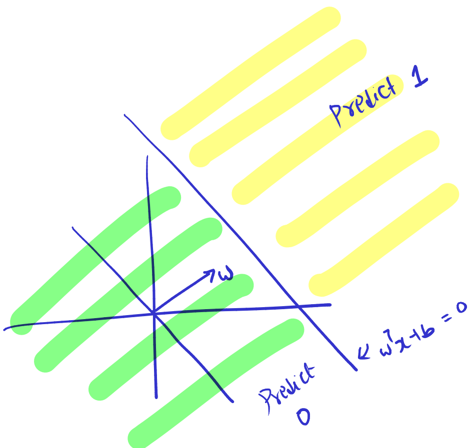
$$e^{-(x_{test}-\hat{\mu}_1)^T \hat{\Sigma}^{-1} (x_{test}-\hat{\mu}_1)} \cdot \hat{p} \geq e^{-(x_{test}-\hat{\mu}_0)^T \hat{\Sigma}^{-1} (x_{test}-\hat{\mu}_0)} (1-\hat{p})$$

On simplification [Take log]

not dependant on data

Predict 1 if $\underbrace{\left((\hat{\mu}_1 - \hat{\mu}_0)^T \hat{\Sigma}^{-1}\right)}_{w^T} x_{test} + \hat{\mu}_0^T \hat{\Sigma}^{-1} \hat{\mu}_0 - \hat{\mu}_1^T \hat{\Sigma}^{-1} \hat{\mu}_1 + \log\left(\frac{1-\hat{p}}{\hat{p}}\right) \geq 0$

$$w^T x_{test} + b \geq 0$$



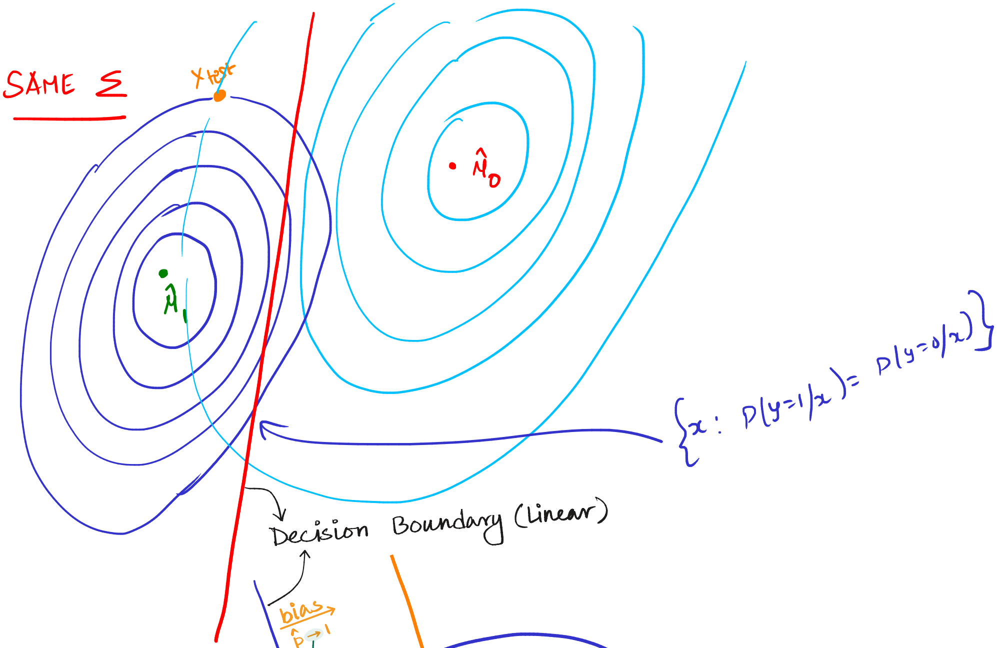
DECISION FUNCTION is LINEAR!

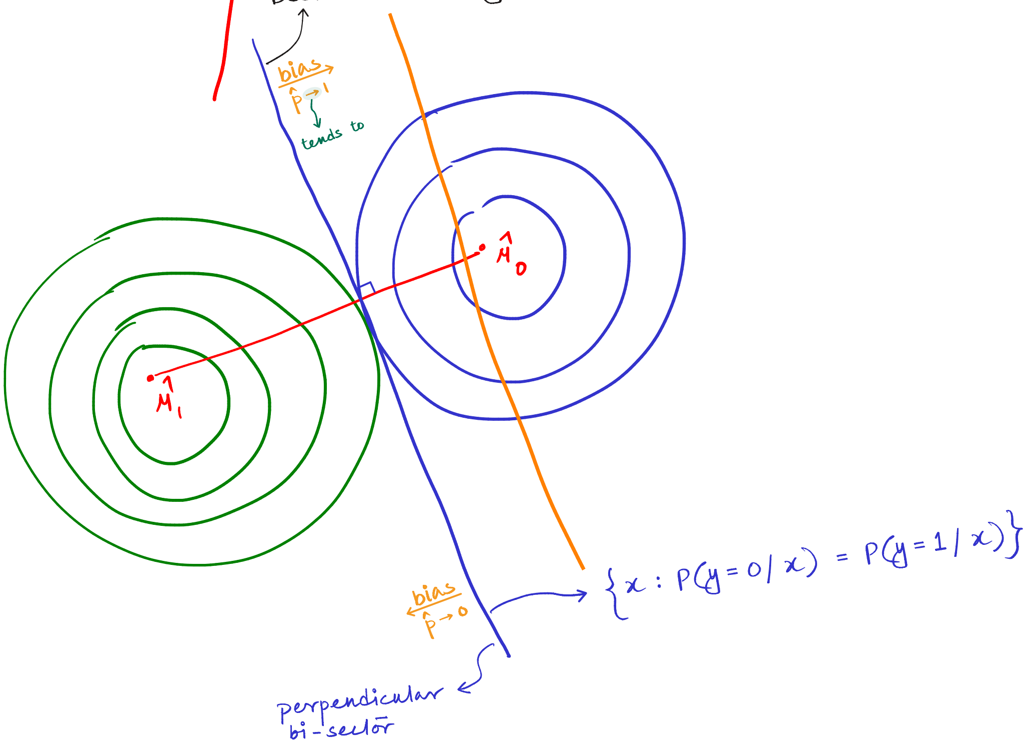
$\hookrightarrow$ * Σ is same for both classes.

SAME Σ

similar type of contours

- elliptical
- directions of the densities would be the same



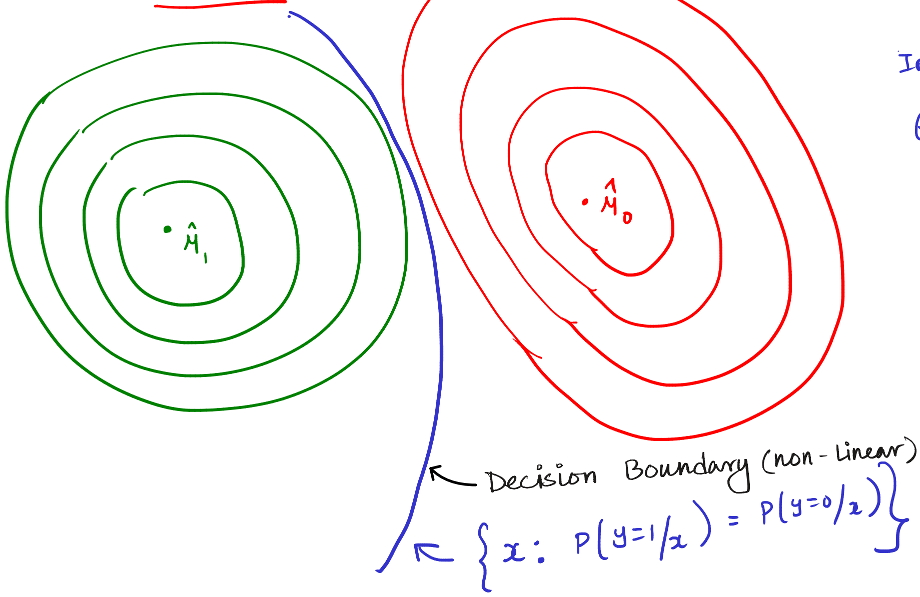


Co-variance matrices are Identity matrices

contours will be circular

* co-variance matrix, common for both classes $\xrightarrow{\text{implies}}$ "Linear decision boundary"

DIFFERENT Σ



In general, Quadratic decision boundary.

GAUSSIAN NAIVE BAYES